

IT25102813 Abivarsan Y.

OOAD Lab 1

Exercise 1

```
import java.util.Scanner;

public class Welcome{

    public static void main(String[] args){

        Scanner scanner = new Scanner(System.in);

        //Getting input - first name

        System.out.print("Enter your first name ");

        String f_name = scanner.nextLine();

        //Getting input - last name

        System.out.print("Enter your last name ");

        String l_name = scanner.nextLine();

        //printing output

        System.out.printf("Welcome to Second Year %s %s\n", f_name, l_name);

        scanner.close();

    }

}
```

Output

Enter your first name Abivarsan

Enter your last name Yogeswaran

Welcome to Second Year Abivarsan Yogeswaran

Exercise 2

```
import java.util.Scanner;

public class marks{

    public static void main(String[] args){

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter number of students: ");

        int num_of_students = scanner.nextInt();

        int[][] marks_array = new int[num_of_students][3]; //[[1] - Mathematics
                                                    //[[2] - Chemistry
                                                    //[[3] - Physics

        int choice;

        do{

            System.out.print("Subject ID\n1. Mathematics\n2. Chemistry \n3. Physics\n");

            System.out.print("\n-----Menu-----\n");

            System.out.print("1. Add Marks of Student\n");

            System.out.print("2. Update Marks of Student\n");

            System.out.print("3. Get Average of a Subject\n");

            System.out.print("4. Get Average of Student\n");

            System.out.print("5. Get total marks of Student\n");

            System.out.print("6. Exit\n");

            System.out.print("\nEnter your choice: ");

            choice = scanner.nextInt();
```

```
break;
```

case 2:

```
System.out.print("\nEnter Student ID: ");  
student_id = scanner.nextInt() - 1;  
  
System.out.print("\nEnter Subject ID: ");  
subject_id = scanner.nextInt() - 1;  
  
System.out.print("\nEnter New Marks: ");  
marks_array[student_id][subject_id] = scanner.nextInt();  
break;
```

case 3:

```
System.out.print("\nEnter the Subject ID, you want to find the average of: ");  
subject_id = scanner.nextInt() - 1;
```

```
int total_marks = 0;
```

```
for (int i = 0; i < num_of_students; i++){  
    total_marks = total_marks + marks_array[i][subject_id];  
}
```

```
float average_of_subject = (float) total_marks/num_of_students;
```

```
switch(subject_id){
```

```
    case 0:
```

```
        System.out.printf("\nAverage marks of Mathematics is %.2f\n",  
average_of_subject);
```

```
        break;
```

```
    case 1:
```

```
                System.out.printf("\nAverage marks of Chemistry is %.2f\n",
average_of_subject);

                break;
```

```
            case 2:

                System.out.printf("\nAverage marks of Physics is %.2f\n",
average_of_subject);

                break;

            }

        break;
```

case 4:

```
    System.out.print("\nEnter the Student's ID, you want to find the average of: ");
```

```
    student_id = scanner.nextInt() - 1;
```

```
    total_marks = 0;
```

```
    for (int i = 0; i < 3; i++){
```

```
        total_marks = total_marks + marks_array[student_id][i];
```

```
    }
```

```
    float average_of_student = (float) total_marks/3;
```

```
        System.out.printf("\nAverage of student %d is %.2f\n", student_id + 1,
average_of_student);
```

```
        break;
```

case 5:

```
    System.out.print("\nEnter the Student's ID, you want to find the average of: ");
```

```
    student_id = scanner.nextInt() - 1;
```

```

        total_marks = 0;

        for (int i = 0; i < 3; i++){
            total_marks = total_marks + marks_array[student_id][i];
        }

        System.out.printf("\nTotal Marks of Student %d is %d \n", student_id + 1, total_marks);
        break;

    case 6:
        break;

    }
}while (choice !=6 );

    scanner.close();
}
}

```

Output

Enter number of students: 2

Subject ID

1. Mathematics

2. Chemistry

3. Physics

-----Menu-----

1. Add Marks of Student

2. Update Marks of Student

3. Get Average of a Subject
4. Get Average of Student
5. Get total marks of Student
6. Exit

Enter your choice: 1

Enter Student ID: 1

Enter Marks for Mathematics: 56

Enter Marks for Chemistry: 76

Enter Marks for Physics: 65

Subject ID

1. Mathematics
2. Chemistry
3. Physics

-----Menu-----

1. Add Marks of Student
2. Update Marks of Student
3. Get Average of a Subject
4. Get Average of Student
5. Get total marks of Student
6. Exit

Enter your choice: 2

Enter Student ID: 1

Enter Subject ID: 2

Enter New Marks: 70

Subject ID

1. Mathematics

2. Chemistry

3. Physics

-----Menu-----

1. Add Marks of Student

2. Update Marks of Student

3. Get Average of a Subject

4. Get Average of Student

5. Get total marks of Student

6. Exit

Enter your choice: 5

Enter the Student's ID, you want to find the average of: 1

Total Marks of Student 1 is 191

Subject ID

1. Mathematics

2. Chemistry

3. Physics

-----Menu-----

1. Add Marks of Student
2. Update Marks of Student
3. Get Average of a Subject
4. Get Average of Student
5. Get total marks of Student
6. Exit

Enter your choice: 1

Enter Student ID: 2

Enter Marks for Mathematics: 90

Enter Marks for Chemistry: 89

Enter Marks for Physics: 88

Subject ID

1. Mathematics
2. Chemistry
3. Physics

-----Menu-----

1. Add Marks of Student
2. Update Marks of Student
3. Get Average of a Subject
4. Get Average of Student
5. Get total marks of Student
6. Exit

Enter your choice: 4

Enter the Student's ID, you want to find the average of: 2

Average of student 2 is 89.00

Subject ID

1. Mathematics

2. Chemistry

3. Physics

-----Menu-----

1. Add Marks of Student

2. Update Marks of Student

3. Get Average of a Subject

4. Get Average of Student

5. Get total marks of Student

6. Exit

Enter your choice: 2

Enter Student ID: 2

Enter Subject ID: 2

Enter New Marks: 95

Subject ID

1. Mathematics

2. Chemistry

3. Physics

-----Menu-----

1. Add Marks of Student
2. Update Marks of Student
3. Get Average of a Subject
4. Get Average of Student
5. Get total marks of Student
6. Exit

Enter your choice: 4

Enter the Student's ID, you want to find the average of: 2

Average of student 2 is 91.00

Subject ID

1. Mathematics
2. Chemistry
3. Physics

-----Menu-----

1. Add Marks of Student
2. Update Marks of Student
3. Get Average of a Subject
4. Get Average of Student
5. Get total marks of Student
6. Exit

Enter your choice: 6

Exercise 3

case 6: // Display Grades Summary

```
System.out.println("\n----- Grade Summary -----");

System.out.printf("%-12s %-12s %-12s %-12s\n", "Student ID", "Math", "Chemistry", "Physics");

System.out.println("-----");

for (int s = 0; s < num_of_students; s++) {

    System.out.printf("%-12d ", s + 1);

    for (int sub = 0; sub < 3; sub++) {

        int score = marks_array[s][sub];

        String grade;

        if (score >= 90) {

            grade = "Grade A";

        } else if (score >= 80) {

            grade = "Grade B";

        } else if (score >= 70) {

            grade = "Grade C";

        } else if (score >= 60) {

            grade = "Grade D";

        } else {

            grade = "Fail";

        }

        System.out.printf("%-12s \n", grade);

    }

    System.out.println(); // Move to next student line
```

```
}  
break;
```

Output

Enter number of students: 1

Subject ID

1. Mathematics

2. Chemistry

3. Physics

-----Menu-----

1. Add Marks of Student

2. Update Marks of Student

3. Get Average of a Subject

4. Get Average of Student

5. Get total marks of Student

6. Table of Students and marks

7. Exit

Enter your choice: 1

Enter Student ID: 1

Enter Marks for Mathematics: 56

Enter Marks for Chemistry: 67

Enter Marks for Physics: 89

Subject ID

1. Mathematics

2. Chemistry

3. Physics

-----Menu-----

1. Add Marks of Student

2. Update Marks of Student

3. Get Average of a Subject

4. Get Average of Student

5. Get total marks of Student

6. Table of Students and marks

7. Exit

Enter your choice: 6

----- Grade Summary -----

Student ID	Math	Chemistry	Physics
------------	------	-----------	---------

1	Fail	Grade D	Grade B
---	------	---------	---------